Image processing and its application – 22RI63 Question Bank

MODULE 1

1. Explain the fundamental steps in Image Processing (10 m)
2. Explain the components of an Image Processing System (10 m)
3. Explain image sensing & acquisition process (10 m)
4. Explain the applications of Image Processing (10 m)
5. Explain different colour models in Image Processing (5 m)
6. Explain how an image is represented (5 m)
7. Explain how Spatial & Grey level resolution affects image quality (5 m)

MODULE 2

1. Explain the basic grey level transforms performed on an image
2. Explain the importance of histogram equilization in IP
3. Explain the working of smoothing frequency domain filters
4. Explain the working of sharpening frequency domain filters
5. Explain the process of image degradation & restoration
6. Explain the different noise models in IP
7. Explain different image filtering techniques
8. Explain the types & importance of image segmentation
9. Explain the techniques in image segmentation
10. Explain image morphological operations

MODULE 3

1. Identify the different types of hyperparameters used in deep learning
2. Illustrate the activation function used in deep learning
3. Explain the effect of batch size on training dataset and types of batch size
4. Apply softmax activation function for input $x = [2,1,0.1]$ and explain its working
5. Identify the parameters affecting overfitting and underfitting condition
6. Interpret the types of loss functions in deep learning
7. Elucidate the different deep learning libraries used in deep learning
8. Summarize the fundamentals of deep learning

MODULE 4

1. Explain the key components of a convolution neural network
2. Illustrate the architecture of LeNet 5
3. Illustrate the architecture of AlexNet
4. Illustrate the architecture of VGG 16
5. Explain two stage object detection method

MODULE 5

1. Explain the working of YOLO algorithm
2. Illustrate the architecture of YOLO network
3. Explain the working of Single shot multibox detector – SSD
4. Explain how skip connections affect neural network with ResNet architecture
5. Explain inception module and dimensionality reduction on GoogLeNet architecture
6. Explain Frame-to-frame motion estimation in visual odometry
7. Explain working and types of Simultaneous Localization and Mapping (SLAM)
8. Explain the challenges in Simultaneous Localization and Mapping (SLAM)